

9. Charlottetown

Shoreline length: 25 km



Shore Types

Charlottetown is a highly developed area located along the Hillsborough River, with a majority of sloped armour shore type.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
25	3	42	0	22	8	0	0

Governance, Infrastructure, and Economy

This area is located entirely within the City of Charlottetown.

Key infrastructure in Charlottetown is concentrated in several geographic areas.

In the downtown and waterfront area, key assets include the Maritime Electric substation on Cumberland Street at the end of Richmond Street; the provincial government offices near Terry Fox Drive and Government Pond; the Prince Edward Island Regiment on Haviland Street; HMCS Queen Charlotte on Water Street; the Prince Edward Battery; Victoria Park Roadway; the downtown waterfront from Government Pond to the PEI Energy Systems District Heating Plant; the Charlottetown Pollution Control Plant; Confederation Landing Park; True North Salmon Co.; Port Charlottetown and the Charlottetown Cruise Ship Terminal; Peake's Quay, including the marina; the Charlottetown Yacht Club; Queen's Wharf and associated marina businesses; the Delta Hotel; Irving Oil; the Charlottetown Event Grounds; Lawton's Drugstore; the Charlottetown boardwalk; and other commercial, industrial, tourism, and institutional development in flood hazard areas or along the shoreline. This coastal area is densely developed, with both large and small lots.

In western Charlottetown, key infrastructure includes Beach Grove Road causeway at Ellen's Creek; North River Road at Heritage Creek and Buchanan Drive; Capital Drive at North River Road and Lower Malpeque Road; Lower Malpeque Road at Oldmore Court; Hermitage Creek Trail; the 36 Signal Regiment Brighton Compound on Brighton Road; and the Trans-Canada Highway at Belvedere Golf Club.

In central and north-central Charlottetown, institutional and health-related assets include Queen Elizabeth Hospital and related health facilities accessed by Murchinson Lane, Hillsborough Hospital, St. Jean Elementary School, École Française Buote, and Holland College.

In eastern and northeastern Charlottetown, key infrastructure includes East Royalty Park, Acadian Drive to the East Royalty Landfill, the Deacon Grove Lane industrial area, and the Confederation Trail.



Queen Elizabeth Hospital



Cruise ship terminal and downtown waterfront



Victoria Park Roadway

Natural Assets and Heritage

Charlottetown is home to Ellens Creek Estuary Natural Area.

There are many municipal parks, greenspaces, sports fields, and boardwalks close to the shoreline, along with several historic sites. These include Andrews Pond Natural Area (hiking) and Wrights Creek Watershed Natural Area (hiking); École Français-Buote's turf field, ballfield, gymnasium, and playground; the Charlottetown Boardwalk; Confederation Park Landing (park, inline skating oval); Victoria Park (tennis and pickleball courts, playground, trails, greenspace, ballfields, soccer fields); Queen Elizabeth Park (tennis and basketball courts, ballfields, soccer/rugby fields); Hermitage Creek Natural Area; Charlottetown Rural High fields and gymnasium; Lewis Point Park (playground, basketball courts); Beaconsfield Historic House (museum); Government Pond, Government House, the Prince Edward Island Battery, the Brighton Beach Range Front and Rear Range Lights, the Charlottetown Event Grounds (also known as Joseph A. Ghiz Memorial Park) and sculpture; Dundas Terrace (a national historic site).

Cultural assets include the Beaconsfield Historic House, Confederation Centre of the Arts, Charlottetown Library Learning Centre, Peakes Quay, Peakes Quay and Harbourfront, and Eastlink Centre.

Park and recreation key infrastructure includes Victoria Park and Victoria Park Boardwalk, Provincial House National Historic Site, Josph A. Ghiz Memorial Park, and Ardgowan National historic Site.

Planning and Policy Framework

Charlottetown has municipal planning authority. Building setbacks are measured from the edge of the buffer, with some exceptions. Some vertical setbacks apply in tidal waterfronts. A draft bylaw currently under development would add further guidance to managing development in hazard areas.

Sediment transport

Sediment transport rates and associated downdrift risk across the SMU are low.

Coastal Flood and Erosion Risk

Coastal flooding is the prevailing hazard for the area. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	108 [4/km]	168 [7/km]	222 [9/km]	322 [13/km]	0	3 [<1/km]	25 [1/km]
Roads	3985 m [163 m/km]	5577 m [229 m/km]	7169 m [294 m/km]	10916 m [447 m/km]	0	0	0

Proposed SMP Strategies

A Resist is appropriate for dense and/or high value infrastructure.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
9.00 Charlottetown	R	R	R
Rationale	Resist strategy is warranted for key infrastructure and densely developed areas, and should be coupled with flood mitigation where necessary.		
9.01 Queen Elizabeth Park -PU	NAI	NAI	MR
Rationale	There are no built assets presently at risk, and a strategy of No Active Intervention would preserve the natural shoreline and beach access. In the long-term, the sports fields to the north may become more vulnerable to coastal flooding, and managed realignment strategies should be explored depending on risk tolerance.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.